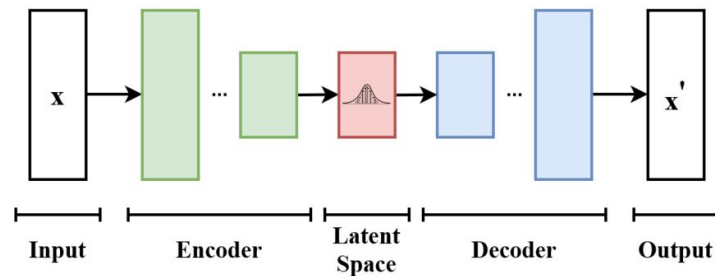
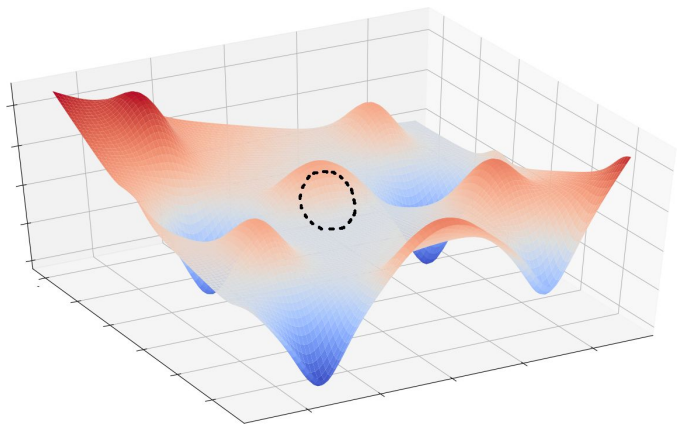


LatentCraft

Property optimization via latent editing

theoretical background

Latent space is a compressed, continuous representation of objects that still retains key information about their manifestations and properties.

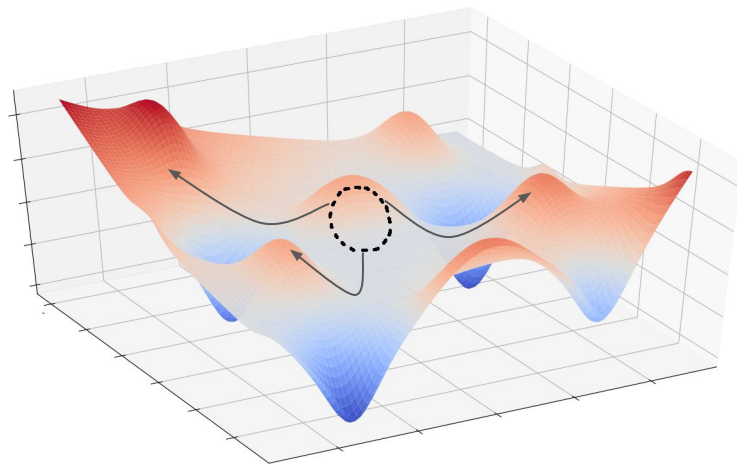


Due to the smoothness of space, **directed movement** in it leads to predictable changes in the properties of decoded objects.

key concept

- 1 Build a continuous latent space that compresses discrete chemical structures
- 2 Pick a seed molecule and map it to a latent vector
- 3 Navigate the latent space toward regions with improved target properties
- 4 Apply uncertainty control to stay in valid regions

Result: novel molecules — optimized and structurally close to the seed



project's goal: develop a latent-space-based optimization pipeline that generates novel molecules with improved target properties while preserving structural similarity to a given seed compound.

applied architectures

- > VAE
- > Autoregressive models
- > Diffusion models

optimization approaches

- Bayesian Optimization <
- Gradient Optimization <
- Evolutionary Algorithms <
- Reinforcement Learning <

existing solutions

chemical_vae

Public

Code for 10.1021/acscentsci.7b00572, now running on Keras 2.0 and Tensorflow

Python 557 190

uncertainty_guided_optimization

Public

Official repository for the paper "Improving black-box optimization in VAE latent space using decoder uncertainty" (Pascal Notin, José Miguel Hernández-Lobato, Yarin Gal)

Python 30 6

lolbo

Public

Python 33 15

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arXiv:2603.26889v1 [cs.LG] 27 Mar 2026

Property-Guided Molecular Generation and Optimization via Latent Flows

Alexander Arjun Lobo , Urvi Awasthi¹ , & Leonid Zhukov
BCG X AI Science Institute

Article | [Open access](#) | Published: 16 February 2026

Enabling multi-target drug discovery through latent evolutionary optimization and synthesis-aware prioritization (EVOSYNTH)

[Viet Thanh Duy Nguyen](#), [Phuc Pham](#) & [Truong-Son Hy](#) ✉

[Communications Chemistry](#) **9**, Article number: 133 (2026) | [Cite this article](#)

4323 Accesses | **2** Altmetric | [Metrics](#)

materials and methods and tools

 PyTorch



Cursor AI



DEAP

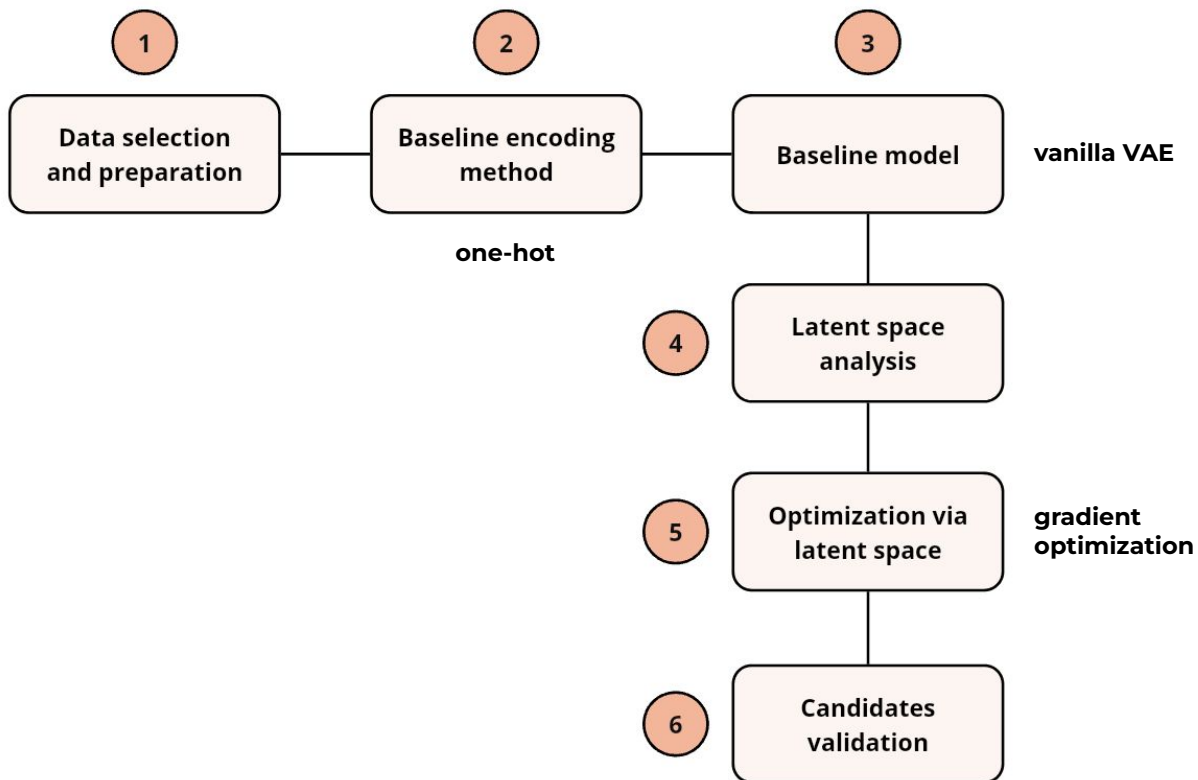
matplotlib



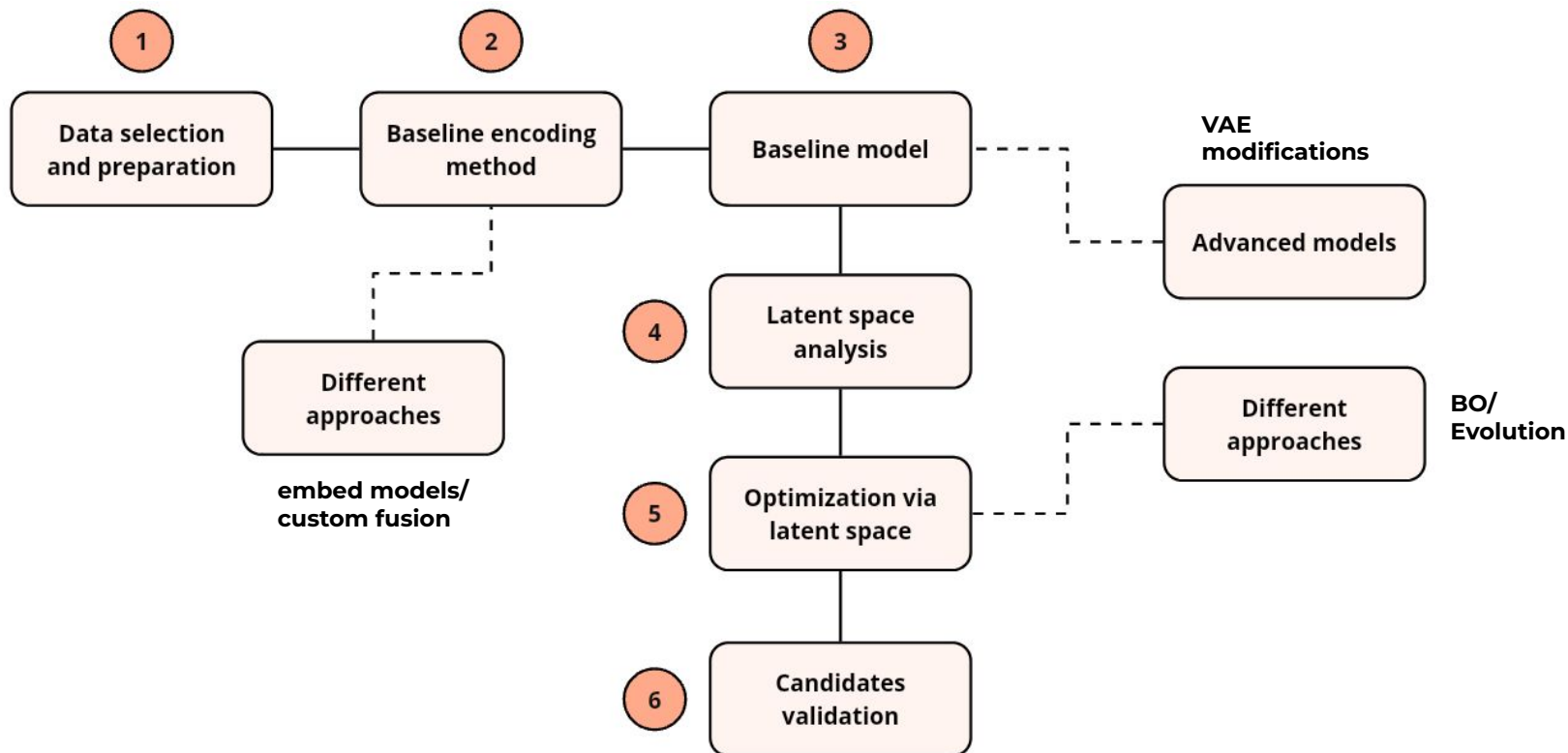
Open-Source Cheminformatics
and Machine Learning

 pandas

baseline workflow



he baseline workflow



what has been done so far

- ✓ The project repository has been created
- ✓ The project plan has been prepared: planned optimization methods and architecture
- ✓ Several datasets for model training were found

next steps

- > Implement the project baseline
- > Consider increasing the complexity of the pipeline and adding new solutions
- > Develop the project to MVP level: add reproducibility, complete documentation

references

Winter, R., Montanari, F., Steffen, A., Briem, H., Noé, F., & Clevert, D. A. (2019). Efficient multi-objective molecular optimization in a continuous latent space. *Chemical science*, 10(34), 8016-8024.

Maus, N., Jones, H., Moore, J., Kusner, M. J., Bradshaw, J., & Gardner, J. (2022). Local latent space bayesian optimization over structured inputs. *Advances in neural information processing systems*, 35, 34505-34518.